

Simple LOW-COST INVERTER

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There are different types of inverters available in market, but their circuits are generally complex, so their cost is high. Here is a proposed low-cost inverter circuit based on MOSFET IRF250, which can also be used as an emergency light.

The circuit is simple and therefore can be wired even on a breadboard. The author's prototype on a breadboard is shown in Fig. 1 while the circuit diagram is shown in Fig. 2.

The circuit comprises step-down transformers X1 and X2, bridge rectifier BR1, 5V voltage regulator 7805 (IC1), 5V single-changeover relay RL1, and a few other components. Capacitors C1 through C3, connected across the supply terminals, minimise ripples and other noise signals. Transformer X2 is an ordinary step-down transformer that is used here for the reverse function, that is, to step up the voltage.

The circuit works on 5V DC, which is obtained through the 230V AC primary to 9V, 500mA step-down transformer X1. The 230V AC mains is connected to the primary of X1 at CON1 in the circuit. The transformer's 9V AC secondary is connected to bridge rectifier BR1 for rectification and filtered by capacitor C1. The rectified and filtered voltage goes to IC LM7805 to get regulated 5V DC voltage for the circuit.

In the inverter, this 5V DC is used to drive/energise relay RL1 when 230V

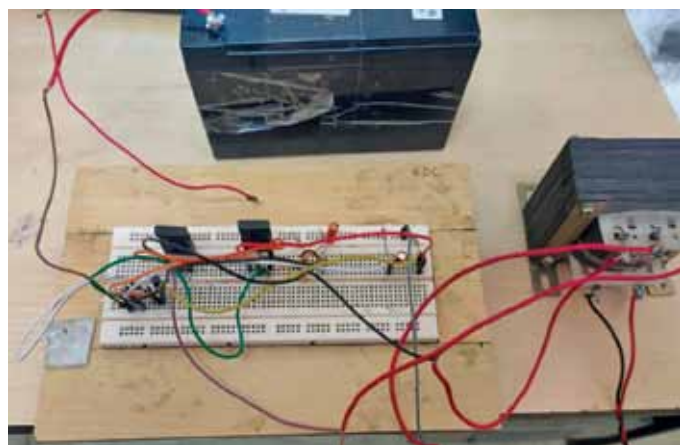


Fig. 1: Author's prototype

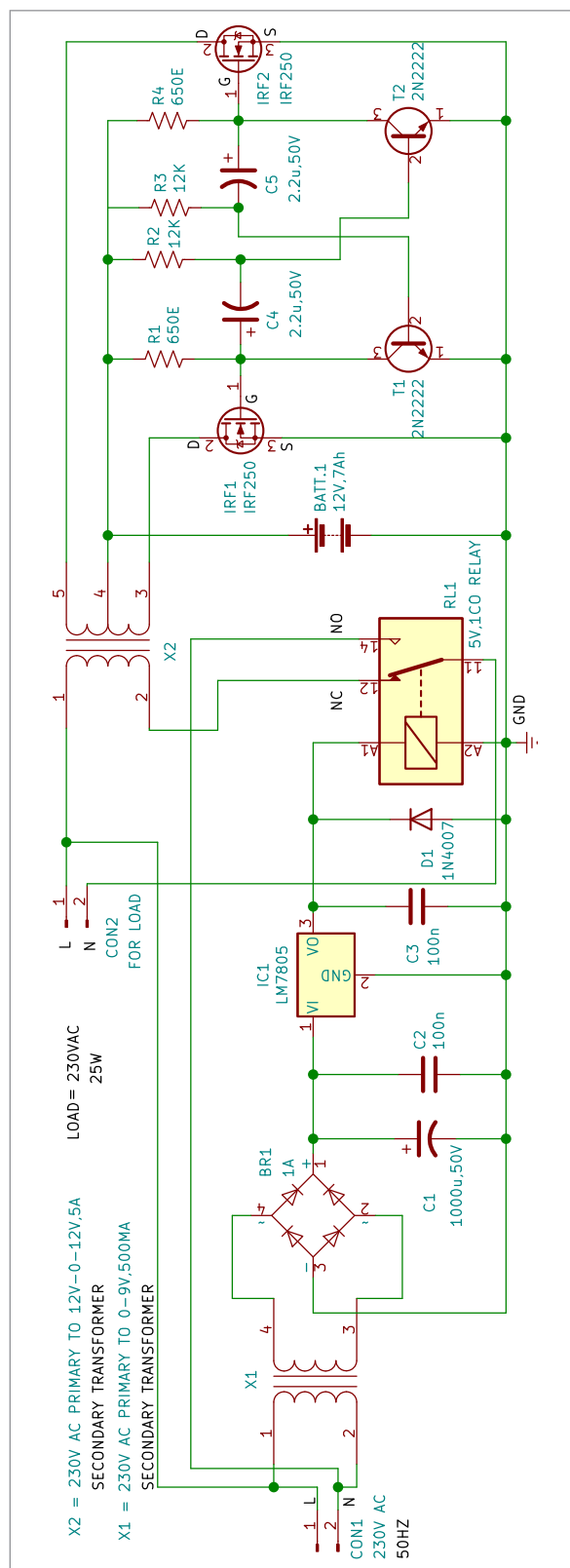


Fig. 2: Circuit diagram of the inverter